A finite-history Lyapunov function at the BDF3 semiconvexity endpoint

Abstract

We consider fully implicit BDF3 for a globally finite, coercive C^1 potential F on a finite-dimensional Euclidean space, where $F + (c/2)\|\cdot\|^2$ is convex and $hc = 95/48$. Every root of each implicit equation is allowed, from an arbitrary initial triple. Pierre's endpoint quadratic identity gives a nonincreasing modified energy with a semidefinite loss. We show that three consecutive zero losses along an actual BDF3 segment force its six points to coincide at a critical point. Summing three modified energies therefore gives an explicit transition-strict Lyapunov function on internally admissible five-point histories. Compactness and closedness of the original relation then imply asymptotic regularity and a compact connected critical omega-limit set at one value of F, for every root selection. The conclusion is critical-set capture; convergence to a single point is not asserted. The comparison with previous work is restricted to the explicitly cited preprint versions.

Keywords: BDF3; semiconvex gradient flow; finite-history Lyapunov function; omega-limit set

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1 Setting and result

Let $m \geq 1$ and let $F : \mathbb{R}^m \to \mathbb{R}$ be continuously differentiable and coercive, meaning that $F(x) \to +\infty$ as $\|x\| \to \infty$. In particular, F is globally finite and attains its minimum $f_* := \min_{\mathbb{R}^m} F$. Fix $c > 0$ such that

$$G(x) := F(x) + \frac{c}{2}\|x\|^2 \quad \text{is convex}, \qquad h > 0, \qquad hc = \frac{95}{48}. \tag{1}$$

Starting from arbitrary $x_0, x_1, x_2 \in \mathbb{R}^m$, we consider the literal BDF3 relation

$$\frac{11}{6}x_j - 3x_{j-1} + \frac{3}{2}x_{j-2} - \frac{1}{3}x_{j-3} = -h\nabla F(x_j), \qquad j \geq 3. \tag{2}$$

An *actual orbit* chooses any root of (2) at each step. There is no minimizing selection or single-valued branch assumption. Write $\operatorname{crit} F = \{q \in \mathbb{R}^m : \nabla F(q) = 0\}$.

Theorem 1. *Under* (1)*, every initial triple admits an infinite actual orbit. Every actual orbit is bounded and satisfies*

$$\|x_n - x_{n-1}\| \longrightarrow 0, \qquad \|\nabla F(x_n)\| \longrightarrow 0, \qquad F(x_n) \longrightarrow F_\infty \in \mathbb{R}. \tag{3}$$

Its position omega-limit set Ω_x is nonempty, compact and connected, and

$$\Omega_x \subset \{q \in \operatorname{crit} F : F(q) = F_\infty\}, \qquad \operatorname{dist}(x_n, \operatorname{crit} F) \longrightarrow 0. \tag{4}$$

For every fixed integer $\ell \geq 1$, the omega-limit set of $(x_{n-\ell+1}, \ldots, x_n)$ is the diagonal copy $\{(q, \ldots, q) : q \in \Omega_x\}$.

The coefficient 95/48 and the quadratic certificate used below are due to Pierre [1, Theorem 2.2]. The modified-energy estimate and the critical omega-limit conclusion in Theorem 3.4 and Corollary 3.5 of that preprint use the strict condition $hc < 95/48$. Endpoint nonincrease, coercive boundedness and summability of the nonnegative losses follow routinely from the same certificate. Our argument supplies the actual-trajectory zero-loss identification at equality, followed by an explicit finite-history function. Summing finitely many energy values is a standard strictification device; the BDF3-specific step is the three-loss rigidity proved in Section 3.

The constant c need not be optimal for the theorem. If a smaller semiconvexity constant $c' < c$ works, then $hc' < 95/48$ and the strict-step result of [1] already applies. Thus the meaningful equality boundary is the case of minimal c.

2 Existence and the source energy

For a triple $(a, b, d) \in (\mathbb{R}^m)^3$, the next root z must satisfy

$$\frac{11}{6}z - 3d + \frac{3}{2}b - \frac{1}{3}a + h\nabla F(z) = 0. \tag{5}$$

The auxiliary function

$$J_{a,b,d}(z) = F(z) + \frac{11}{12h}\|z\|^2 - \frac{1}{h}\langle 3d - \frac{3}{2}b + \frac{1}{3}a, z\rangle$$

is coercive because F is bounded below. It is C^1, so a global minimizer satisfies (5). This proves existence of a next root for every triple and hence recursive existence of an infinite orbit. Minimization is used only in this existence proof; all roots remain allowed.

Set $d_j = x_j - x_{j-1}$ for $j \geq 1$ and, for $j \geq 3$, define

$$B_j = \frac{11}{6}d_j - \frac{7}{6}d_{j-1} + \frac{1}{3}d_{j-2}, \qquad R_j = d_j - \frac{7}{4}d_{j-1} + d_{j-2}. \tag{6}$$

Then (2) is exactly $B_j = -h\nabla F(x_j)$. Pierre's quadratic form is

$$Q(u, v) = \frac{1}{6}\left(\|v - \tfrac{7}{4}u\|^2 + \|u\|^2\right). \tag{7}$$

It is positive definite on $(\mathbb{R}^m)^2$. The scalar identity of [1, Theorem 2.2], applied componentwise, reads

$$\langle \frac{11}{6}u - \frac{7}{6}v + \frac{1}{3}w, u\rangle = Q(u, v) - Q(v, w) + \frac{95}{96}\|u\|^2 + \frac{1}{6}\|u - \tfrac{7}{4}v + w\|^2. \tag{8}$$

Define the oriented convex Bregman loss and the modified energy by

$$D_G(a, b) = G(a) - G(b) - \langle \nabla G(b), a - b\rangle \geq 0, \tag{9}$$

$$V_j = F(x_j) - f_* + h^{-1}Q(d_j, d_{j-1}), \qquad j \geq 2. \tag{10}$$

Expanding (9) at (x_{j-1}, x_j) gives

$$D_G(x_{j-1}, x_j) = F(x_{j-1}) - F(x_j) + \langle \nabla F(x_j), d_j \rangle + \frac{c}{2}\|d_j\|^2.$$

Use $B_j = -h\nabla F(x_j)$, (8) and $hc/2 = 95/96$ to obtain

$$L_j := V_{j-1} - V_j = D_G(x_{j-1}, x_j) + \frac{1}{6h}\|R_j\|^2, \qquad j \geq 3. \tag{11}$$

Thus $(V_j)_{j\geq 2}$ is nonnegative and nonincreasing. In particular,

$$\sum_{j=3}^{\infty} D_G(x_{j-1}, x_j) < \infty, \qquad \sum_{j=3}^{\infty} \|R_j\|^2 < \infty. \tag{12}$$

The residual in (8) is only semidefinite; we will use both terms of (11) and the original recurrence to identify zero loss.

For later compactness arguments, introduce the triple energy

$$E(a, b, d) = F(d) - f_* + h^{-1}Q(d - b, b - a) \tag{13}$$

and the set-valued relation

$$T_3(a, b, d) = \{(b, d, z) : z \text{ satisfies (5)}\}.$$

The function E is continuous, nonnegative and has compact sublevel sets. Indeed, $E \leq M$ bounds d by coercivity and bounds $d - b$ and $b - a$ by positive definiteness of Q, hence bounds the whole triple. The sublevel is closed. For every allowed successor, (11) gives

$$E(b, d, z) \leq E(a, b, d). \tag{14}$$

Consequently every actual orbit is bounded, with no condition on its initial triple.

The graph of T_3 is closed because (5) is a continuous equality. Every value is nonempty and compact: existence was proved above, and (14) confines the value to a compact E-sublevel. More generally, if the input triples range over a compact set C, then $\max_C E < \infty$ bounds all outputs. The graph restricted to C is therefore compact. These facts use the whole root relation.

3 Three consecutive zero losses

We first record the consequence of convexity that the residual alone does not provide. For a differentiable convex function G on $\mathbb{R}^m$,

$$D_G(a, b) = 0 \quad \Longrightarrow \quad \nabla G(a) = \nabla G(b). \tag{15}$$

Indeed, let $p = \nabla G(b)$. The supporting inequality $G(y) \geq G(b) + \langle p, y - b\rangle$ and equality at a imply $G(y) \geq G(a) + \langle p, y - a\rangle$ for every y. Thus p is a convex subgradient at a, and differentiability identifies it with $\nabla G(a)$.

Write $p_j = \nabla G(x_j)$. Subtracting two consecutive equations (2) and using (1) yields, whenever both steps are present,

$$\begin{aligned} h(p_j - p_{j-1}) &= \frac{7}{48}d_j + 3d_{j-1} - \frac{3}{2}d_{j-2} + \frac{1}{3}d_{j-3} \\ &= \frac{17}{16}(d_j + d_{j-1}) - \frac{11}{12}R_j + \frac{1}{3}R_{j-1}. \end{aligned} \tag{16}$$

The second line is an exact rearrangement. This identity retains the original BDF3 equations; convexifying the potential has not changed the dynamics.

Lemma 1 (Three-loss rigidity)**.** *Let $k \geq 3$. Suppose six successive points $x_{k-3}, \ldots, x_{k+2}$ satisfy (2) at $k, k+1, k+2$. Then $L_k = L_{k+1} = L_{k+2} = 0$ if and only if all six points coincide at some $q \in \operatorname{crit} F$.*

Proof. Zero loss in (11) gives $R_k = R_{k+1} = R_{k+2} = 0$ and, by (15), $p_{k-1} = p_k = p_{k+1} = p_{k+2}$. Apply (16) at $k+1$ and $k+2$ to obtain

$$d_{k+1} = -d_k, \qquad d_{k+2} = -d_{k+1} = d_k.$$

It follows that

$$0 = R_{k+2} = d_{k+2} - \frac{7}{4}d_{k+1} + d_k = \frac{15}{4}d_k.$$

Thus $d_k = d_{k+1} = d_{k+2} = 0$. Now $R_{k+1} = 0$ gives $d_{k-1} = 0$, and $R_k = 0$ gives $d_{k-2} = 0$. These five increments join all six points, so all are equal to a point q. Equation (2) at k gives $\nabla F(q) = 0$. The converse follows by substitution. ∎

In particular, every complete actual trajectory with zero loss at every step is constant at a critical point. The finite-block statement itself does not require boundedness.

4 The admissible five-point energy

For $H = (z_0, z_1, z_2, z_3, z_4) \in (\mathbb{R}^m)^5$, define

$$\mathcal{A}_5 = \left\{ H : \frac{11}{6}z_i - 3z_{i-1} + \frac{3}{2}z_{i-2} - \frac{1}{3}z_{i-3} + h\nabla F(z_i) = 0, \quad i = 3, 4 \right\}. \tag{17}$$

Only these two internal equations are imposed. For $H \in \mathcal{A}_5$, let $T_5(H)$ contain every shifted history $H' = (z_1, z_2, z_3, z_4, z_5)$ satisfying

$$\frac{11}{6}z_5 - 3z_4 + \frac{3}{2}z_3 - \frac{1}{3}z_2 + h\nabla F(z_5) = 0. \tag{18}$$

Every such H' belongs to $\mathcal{A}_5$. Define

$$W(H) = E(z_0, z_1, z_2) + E(z_1, z_2, z_3) + E(z_2, z_3, z_4). \tag{19}$$

Theorem 2 (Transition-strict finite-history function)**.** *The domain $\mathcal{A}_5$ is closed, and T_5 has nonempty compact values and a closed graph. The function W is continuous and nonnegative, with compact sublevel sets on $\mathcal{A}_5$, each invariant under every allowed successor. For every $H' \in T_5(H)$,*

$$W(H') \le W(H), \qquad W(H') = W(H) \iff z_0 = z_1 = \cdots = z_5 = q \in \operatorname{crit} F. \tag{20}$$

The fixed phase set $\{H : H \in T_5(H)\}$ is

$$\mathcal{E}_5 = \{(q, q, q, q, q) : q \in \operatorname{crit} F\}.$$

Every original orbit enters $\mathcal{A}_5$ at $H_4 = (x_0, x_1, x_2, x_3, x_4)$, after precisely two new implicit steps. For $H_n = (x_{n-4}, \ldots, x_n)$, the lifted transitions start at $H_4 \to H_5$ and satisfy

$$W(H_{n-1}) - W(H_n) = L_{n-2} + L_{n-1} + L_n, \qquad n \ge 5. \tag{21}$$

Proof. The defining equations are continuous, so the domain and graph are closed. Existence in Section 2 gives a successor for every input. The formula for W is continuous and nonnegative. A bound on W bounds its first and third terms separately; properness of E then bounds all five coordinates. Thus its sublevels, on the whole five-point space and on the closed subset $\mathcal{A}_5$, are compact.

For a transition through $z_0, \ldots, z_5$, write $V_i = E(z_{i-2}, z_{i-1}, z_i)$, $i = 2, 3, 4, 5$, and $L_i = V_{i-1} - V_i$, $i = 3, 4, 5$. The two internal equations and the new equation make (11) valid at all three indices. Hence

$$W(H) - W(H') = V_2 - V_5 = L_3 + L_4 + L_5 \ge 0. \tag{22}$$

Equality is equivalent to three zero losses, so Lemma 1 proves (20). Sublevels are invariant; a closed value confined to such a sublevel is compact. A fixed phase must have all its coordinates equal, and its recurrence then makes that common point critical. Conversely, a critical diagonal has the stationary self-transition.

For arbitrary x_0, x_1, x_2, the equations at x_3, x_4 are exactly those needed for $H_4 \in \mathcal{A}_5$. No equations are added at the original three points. Finally, $W(H_n) = V_{n-2} + V_{n-1} + V_n$ for $n \ge 4$, and subtraction gives (21). ∎

Remark. Strictness is a property of the six-point transition. Every transition from $\mathcal{A}_5 \setminus \mathcal{E}_5$ strictly decreases W. At a critical diagonal, only the stationary self-transition has zero drop; any different admissible successor has strictly lower energy. The theorem does not assert that every continuation from a critical diagonal stays there, nor that an individual critical phase is stable.

5 Compact limits and asymptotic regularity

Proof of Theorem 1. Existence and boundedness were proved in Section 2. Lift an arbitrary actual orbit to $H_n \in \mathcal{A}_5$, $n \geq 4$. Theorem 2 gives the compact invariant set

$$K = \{H \in \mathcal{A}_5 : W(H) \leq W(H_4)\},$$

and $W(H_n)$ converges to a finite value w_∞. The lifted omega-limit set Ω_5 is nonempty and compact. If $H_{n_j} \to H_* \in \Omega_5$, compactness gives, after a further subsequence, $H_{n_j+1} \to H_+ \in \Omega_5$. The closed graph implies $H_+ \in T_5(H_*)$, while continuity gives

$$W(H_*) = w_\infty = W(H_+).$$

By (20), both are the same critical diagonal. Therefore

$$\Omega_5 \subset \mathcal{E}_5. \tag{23}$$

This uses a limit of actual successors, not an assumption that every successor of a limit phase belongs to the omega-limit set.

For completeness, all fixed integer shifts can be extracted at once. Starting with $H_{n_j} \to H_*$, successively extract subsequences for $r = 1, -1, 2, -2, \ldots$ and take the diagonal subsequence. For every fixed $r \in \mathbb{Z}$, the indices $n_j + r$ are eventually at least 4 and the histories lie in K, so

$$H_{n_j+r} \longrightarrow \widehat{H}_r \in \Omega_5.$$

Passing the consecutive relations through the closed graph gives $\widehat{H}_{r+1} \in T_5(\widehat{H}_r)$ for all $r \in \mathbb{Z}$. Thus there is a two-sided complete limit trajectory through $\widehat{H}_0 = H_*$, on which $W \equiv w_\infty$. Every edge is a stationary critical self-transition by (20). No single-valued branch or abstract invariance theorem is needed.

Compactness and (23) imply $\operatorname{dist}(H_n, \mathcal{E}_5) \to 0$: otherwise a subsequence staying a positive distance away would have a limit in Ω_5. They also imply $\|d_n\| \to 0$, since a subsequence with separated last two coordinates would have a nondiagonal limit. The unchanged BDF3 equation then gives

$$\|\nabla F(x_n)\| \leq \frac{1}{h}\left(\frac{11}{6}\|d_n\| + \frac{7}{6}\|d_{n-1}\| + \frac{1}{3}\|d_{n-2}\|\right) \longrightarrow 0.$$

Since $V_n \to v_\infty \geq 0$ and $Q(d_n, d_{n-1}) \to 0$, we have $F(x_n) \to f_* + v_\infty =: F_\infty$.

Boundedness makes Ω_x nonempty and compact. Continuity of F and ∇F gives the inclusion in (4). A bounded sequence approaches its own omega-limit set: otherwise a subsequence a fixed positive distance away would have a new limit point. This proves $\operatorname{dist}(x_n, \operatorname{crit} F) \to 0$ as well.

To see connectedness, suppose that $\Omega_x = A \cup B$ for two disjoint nonempty compact sets A, B. Their distance is positive. Choose small neighborhoods of A and B that

are separated by a positive gap. Eventually their union contains every x_n, since the sequence approaches Ω_x, and each neighborhood is visited infinitely often. But $\|x_n - x_{n-1}\| \to 0$ makes steps eventually smaller than the gap, so the sequence cannot switch between the neighborhoods while remaining in their union. This contradiction proves connectedness. Finally, for any fixed ℓ, finitely many consecutive vanishing increments make all coordinates of an ℓ-history asymptotically equal. Taking limits along position subsequences, and projecting limits of history subsequences, proves both inclusions in the stated diagonal omega-limit identity. ∎

6 Source and verification scope

The precise source comparison is with arXiv:2002.03925v1 [1], especially Theorem 2.2 (p. 3), Theorem 3.4 (p. 8) and Corollary 3.5 (p. 9). The inspected high-order preprint [2, Theorem 3.2 and Corollary 3.4] also uses a strict step condition. The Cahn–Hilliard review [3, Section 4.1] concerns additional PDE structure and is not used as a generic endpoint theorem here. The 2021 and 2022 final journal bodies and the review's final revision have not been verified for this comparison. We do not assert preprint/final equivalence, firstness or exhaustive literature coverage.

The result proved here is an endpoint critical-set conclusion for the stated C^1, finite-dimensional, all-root scheme. It does not include single-point or Kurdyka–Łojasiewicz convergence, rates, a nonsmooth extension or a larger step interval. The companion standard-library Python script verifies the displayed polynomial and linear identities using exact rational arithmetic, including the zero-loss system and the finite-history telescoping. It checks algebra only; the convexity, existence, compactness and connectedness arguments are the proofs above.

References

[1] M. Pierre, Maximum time step for the BDF3 scheme applied to gradient flows, preprint, arXiv:2002.03925v1, 10 February 2020. `https://arxiv.org/abs/2002.03925v1`.

[2] M. Pierre, Maximum time step for high order BDF methods applied to gradient flows, preprint, HAL hal-03438159v1, 21 November 2021. `https://hal.science/hal-03438159v1`.

[3] M. Brachet, P. Parnaudeau and M. Pierre, Convergence to equilibrium for time and space discretizations of the Cahn–Hilliard equation, preprint, HAL hal-03466113v1, 4 December 2021. `https://hal.science/hal-03466113v1`.